Tahsin Tajwar Tanni

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[Portfolio](#) | [LinkedIn](#) | [Github](#) | [Google Scholar](#)

PERSONAL PROFILE

- Computer Science and Engineering graduate with a thesis focused on pretrained language model security, specializing in defense mechanisms against fine-tuning attacks and improving model robustness under adversarial conditions.
- Skilled in deep learning research and experimentation using PyTorch and Hugging Face Transformers, full-stack web development with React, Node.js, and MongoDB, and reproducible research workflows in Python.

EDUCATION

01/2022 – 02/2026 *BRAC UNIVERSITY*
BSc in Computer Science and Engineering GPA: 3.60 / 4.00

03/2018 – 11/2020 *Govt. Haji Muhammad Mohsin College*
Higher Secondary Certificate (HSC) GPA: 5.00 / 5.00

01/2004 – 12/2018 *Bangladesh Mohila Samity Girls' High School and College*
Secondary School Certificate (SSC) GPA: 5.00 / 5.00

PROFESSIONAL EXPERIENCE

05/2026 – Present **AI Automation Trainee**
180 RE, Toronto, Canada
- Built LLM-powered workflows using OpenAI and Anthropic APIs, automating multi-step business processes via n8n and third-party service integrations.
- Developed and deployed AI-powered chatbots, integrating them into websites to automate user interactions and supports

02/2023 – 07/2024 **Junior Web Developer**
Weabers, Chattogram, Bangladesh
- Built responsive web applications; collaborated on API integration and feature deployment.

VOLUNTEERING

02/2025 - 07/2025 **Python Instructor**
Zentorra
- Taught foundational Python programming concepts to beginner-level learners.

RESEARCH EXPERIENCE

01/2025 – 01/2026 **Undergraduate Researcher**
BRAC University, Department of Computer Science and Engineering
01/2025-01/2026
- Investigated post-deployment vulnerabilities in pretrained language models, focusing on weight noise injection and low-precision quantization attacks that degrade model stability without altering architecture.
- Designed and implemented Anchor-Guided Repair, a fine-tuning defense that jointly optimizes language modeling loss and anchor regularization to constrain parameter drift from a clean task-adapted baseline.

• Evaluated the method across multiple LLM architectures and attack types (Gaussian noise, low-bit quantization), consistently recovering performance close to the clean baseline.

PUBLICATION

Rohan, A. M., Khan, N., **Tanni, T. T.**, Fardin, F., & Bushra, A. (2026). *Anchor-Guided Repair: A Defense Mechanism for Enhancing Stability of Compromised Pretrained Language Models Against Low-Precision and Weight Noise Attacks*. In *Proceedings of IndabaX Nigeria 2026: Building Scalable AI That Works: From Research to Deployment in Resource-Constrained Environments*. *Proceedings of Machine Learning Research*, **319**, 368–381. https://proceedings.mlr.press/v319/

ONGOING RESEARCH

Phantom Guard: A Fine-tuned Multi-Agent System for Robust Offline Slopsquatting Detection *(Role: Researcher)*

• Designing a dual-LLM generator-reviewer pipeline to detect hallucinated and slopsquatted package recommendations.
• Implementing fallback validation against official package registries and repository APIs.
• Exploring knowledge distillation for lightweight, low-latency self-verifying models.

TECHNICAL SKILLS

• **Programming Languages:** Python, JavaScript, C
• **Machine Learning & AI:** PyTorch, Hugging Face Transformers, scikit-learn, Deep Learning, Natural Language Processing (NLP)
• **Research Methods:** Model Fine-tuning, Transfer Learning, Adversarial Machine Learning, Performance Benchmarking, Reproducible Research
• **Scientific Computing:** NumPy, pandas
• **Tools & Platforms:** Docker, Git, GitHub, Hugging Face Spaces, LaTeX
• **Software Development:** React.js, Node.js, Express.js, MongoDB, MySQL, SQLite

LANGUAGE PROFICIENCY

• **English:** Full professional proficiency - Medium of Instruction (MOI) certified, BRAC University, 2026
• **Bengali:** Native

AWARDS & HONORS

• **Dean's List, BRAC University**, Fall 2024
• **Vice Chancellor's List, BRAC University**, Summer 2025
• **2nd Runner-up - CSE471 Project Demonstration**, BRAC University, 2025
• **Academic Distinction**, BSc in Computer Science and Engineering, BRAC University, 2026

REFERENCES

Dr. Muhammad Iqbal Hossain
Thesis Supervisor
Associate Professor,
Department of CSE
BRAC University
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Dr. Md. Sakir Hossain
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